

Supporting Information for : A matter of size: what underpinned the emergence and demise of the largest rodents that ever lived?

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This file contains:

- Supplementary text
- Supplementary figures
- Supplementary tables

Other Supplementary Materials may be found at <placeholder>.

Codes are available at: <https://github.com/Buffer3369/Chinchilloids.git>

Supplementary text

Some details regarding decisions we took during our occurrence data cleaning

Age ranges of the Miocene occurrences from the Río Santa Cruz (Santacrucian SALMA) were sharpened based on their assigned locality (Arnal et al. 2019). Hence, occurrence from Barrancas Blancas were respectively assigned the 16.3–17.21 and 15.3–16.47 Ma intervals.

Given that *Lagostomus* (*Lagostomopsis*) is a subgenus (Rasia and Candela 2012), we set the **genus** column of each *Lagostomus* (*Lagostomopsis*) occurrence to *Lagostomus*, and switched their **genus_level_status** from **extinct** to **extant**.

All specimens referred to as *Lagostomus minimus*, *L. cavifrons*, *L. heterogenidens*, *L. debilis* and *L. egenus* were synonymised with *Lagostomus maximus*, following (Ubilla and Rinderknecht 2016; Rasia 2024).

The single occurrence assigned to *Lagostomus striatus* was removed as considered *nomen nudum* by Rasia (2016).

Uncertain age of the Ituzáingo formation was set to Huayquerian SALMA (Late Miocene), *ca.* 5–8 Ma.

Age range of occurrences from Bucher et al. (2021) were set to the one of the original publication: 14.86–12.75 Ma when coming from the “Cruces Infinitos” and “Los Yeguarizos localities”, 12.75–12 Ma when coming from the “La Gloria” and “La Hoyada” localities.

Based on Brandoni et al. (2016), the age range of the occurrences from the Pinturas fauna (*e.g.*, (Kramarz 2002; 2004; 2006)), Burdigalian, were set between 20 and 16.8 Ma.

According to (Esteban et al. 2014; Esteban et al. 2019), the age range of the occurrences belonging to the Andalhualá formation was set to Huayquerian–Chapadmalalan (*c.* 7.14–3.66 Ma).

Perimys perpinguis and *Perimys impactus* were lumped in *Perimys onostus* according to McGrath et al. (2022).

Age range of the occurrence belonging to the Sierra Baguales (Santa Cruz formation, Magallanes, Chile) were set to 19–17.8 Ma according to the isotopic constraints provided in Bostelmann et al. (2013).

Age range of the occurrences reported from the Cerro Boleadoras Formation was set to 16.5–15.1 Ma, according to Vizcaino et al. (2022).

Added two occurrences of *Saremmys ligcura* from the two localities of Bryn Gwyn and Gran Barranca (Chubut, Argentina), according to Busker et al. (2019).

We acknowledge the proposition of synonymisation between *Telicomys gigantissimus* and *Telicomys giganteus* made by Rasia et al. (2025). However, in the light of phylogenetic analyses that are currently in progress, we decided not to account for it, therefore treating *T. gigantissimus* and *T. giganteus* as two distinct taxa.

Additional details are provided in the **Remark** column of *Dataset S1*.

Supplementary Figures

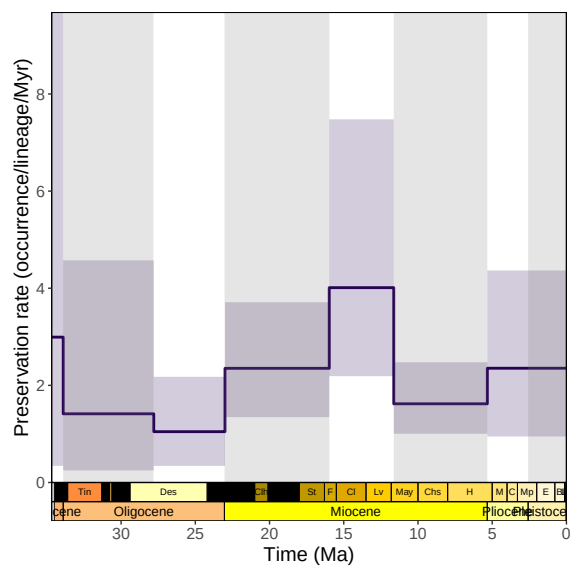


Figure S 1. Preservation rate through time in our species-level dataset estimated with PyRate. The solid line represents the mean posterior estimate and the shaded area the associated 95% HPD interval.

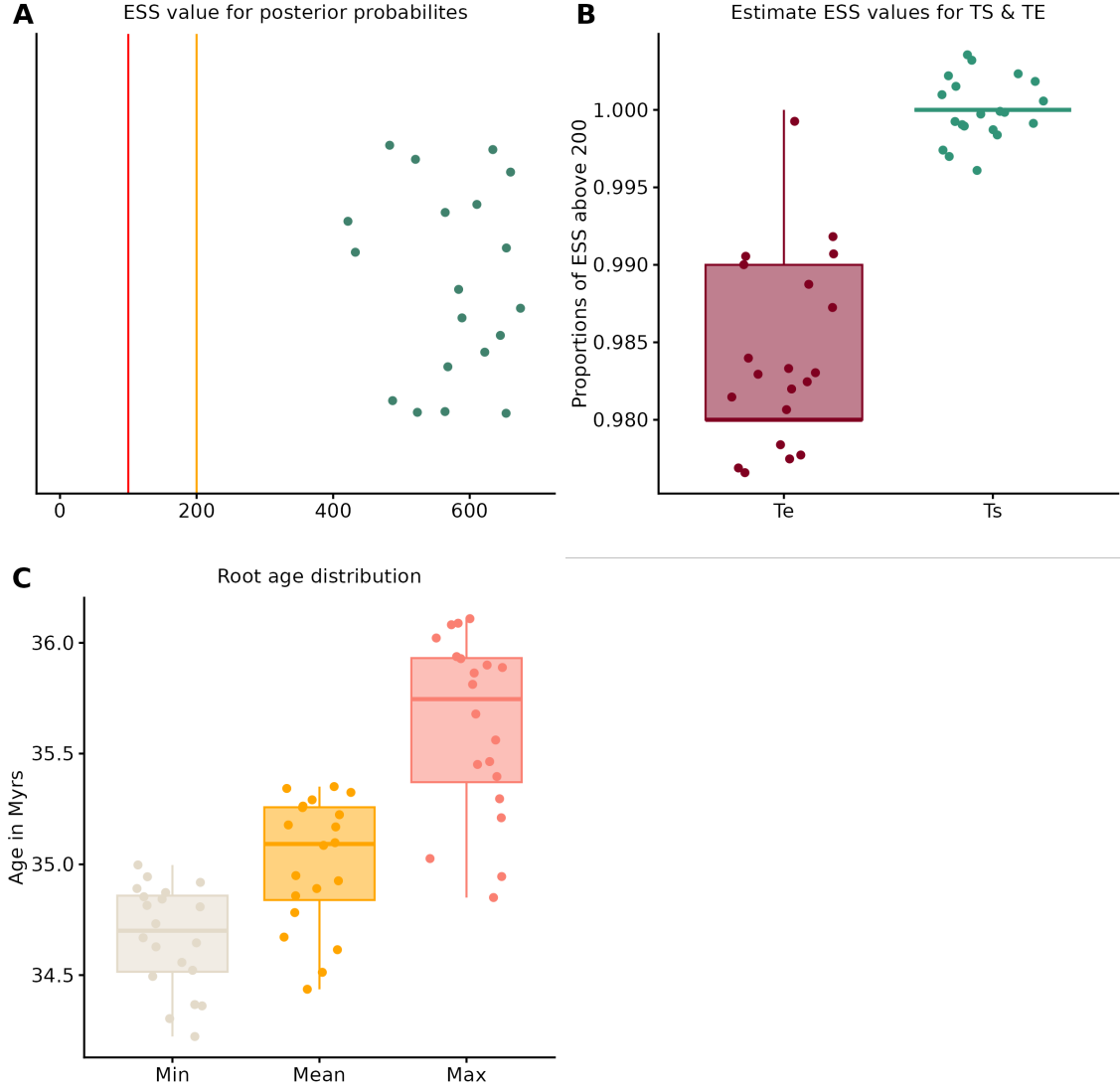


Figure S 2. Convergence of the PyRate runs for the analysis of the species-level temporal occurrence database. The plot (A) the ESS of the model posterior probability for all the 20 replicates we ran. ESS greater than 200 (first-scale indicator of a run that converged, $n = 20$ in that case) are displayed in green. The proportion of origination and extinction times (resp. T_S and T_E) that converged (i.e., with $ESS > 200$) are shown in (B), for each of our 20 replicates. The plot (C) shows the posterior estimates of the mean root age of our group, together with their lower and upper boundaries of the 95% HPD interval, across the 20 replicates.

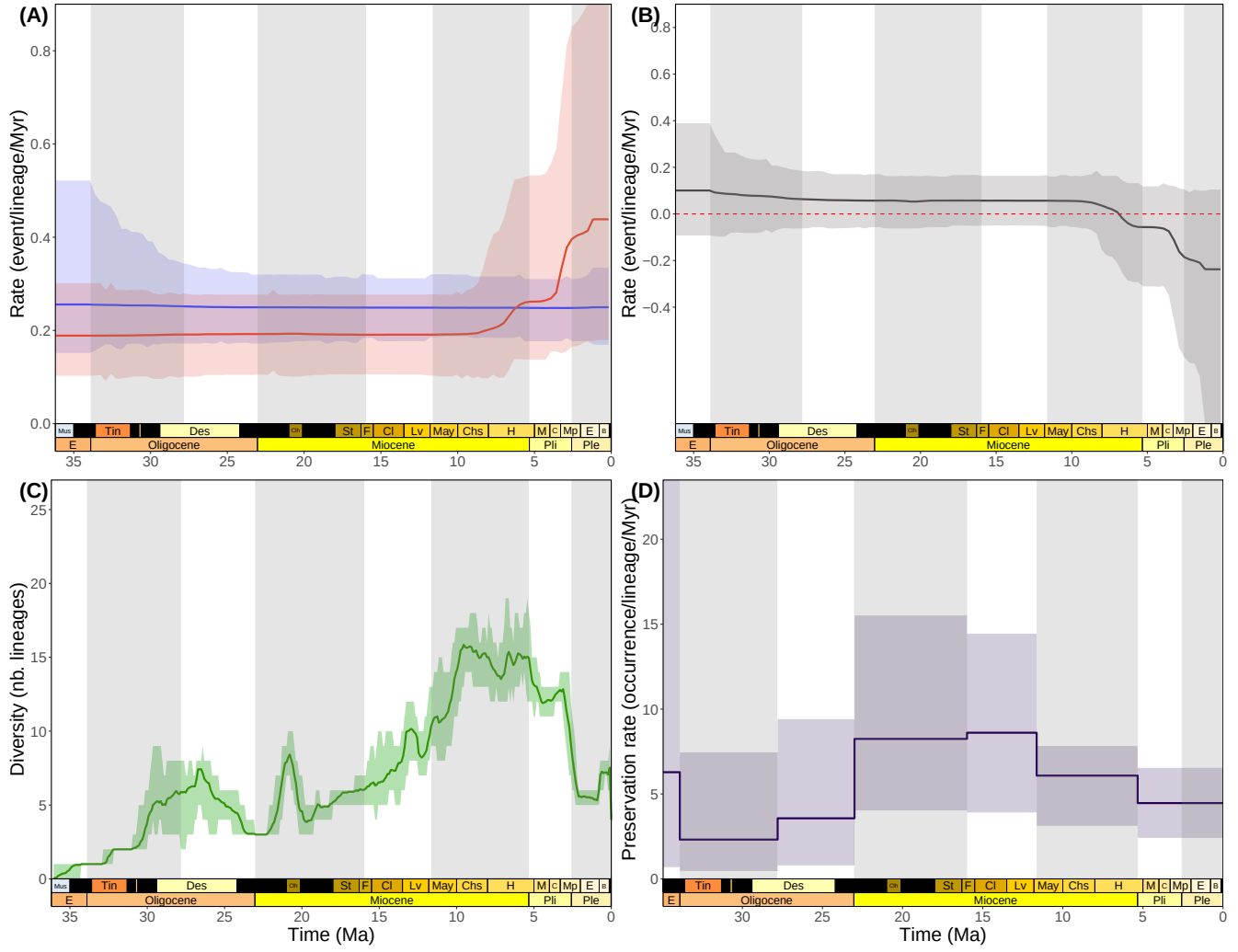


Figure S 3. Origination and extinction (A), net diversification (= origination - extinction) (B) rates, diversity (C) and preservation rate (D) through time as estimated by the analysis of our genus-level occurrence dataset with the rjMCMC model.

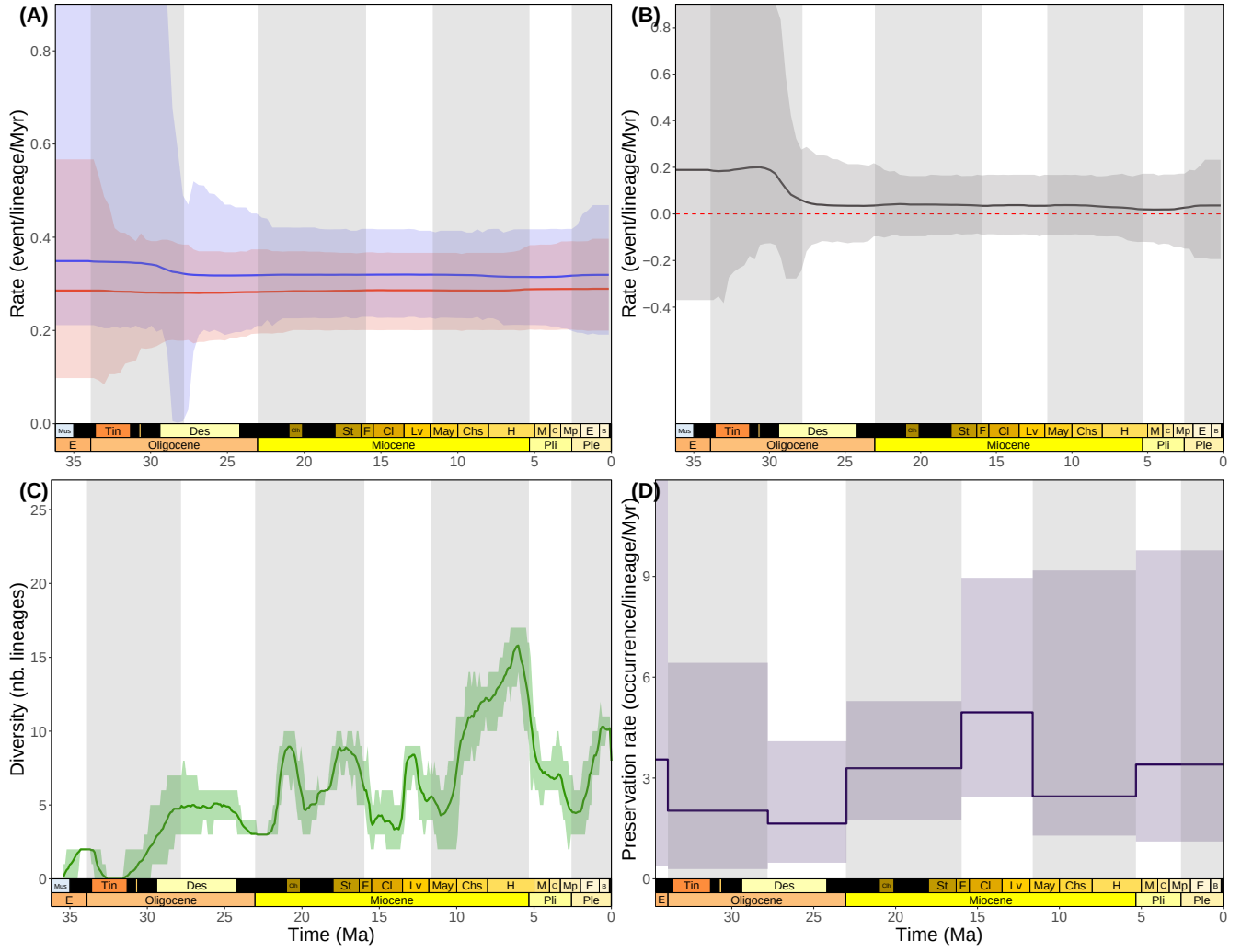


Figure S 4. Speciation and extinction (A), net diversification (= origination - extinction) (B) rates, diversity (C) and preservation rate (D) through time as estimated by the analysis of our species-level occurrence dataset with the rjMCMC model after removing singletons.

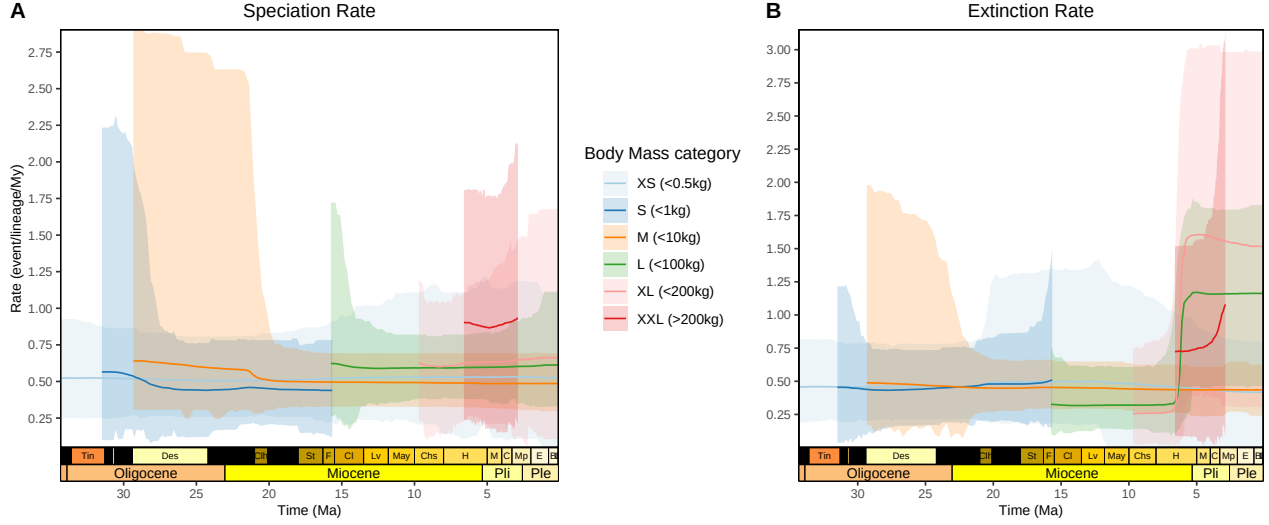


Figure S 5. Speciation (A) and extinction (B) rates through time per body mass class as inferred by the RJMCMC model.

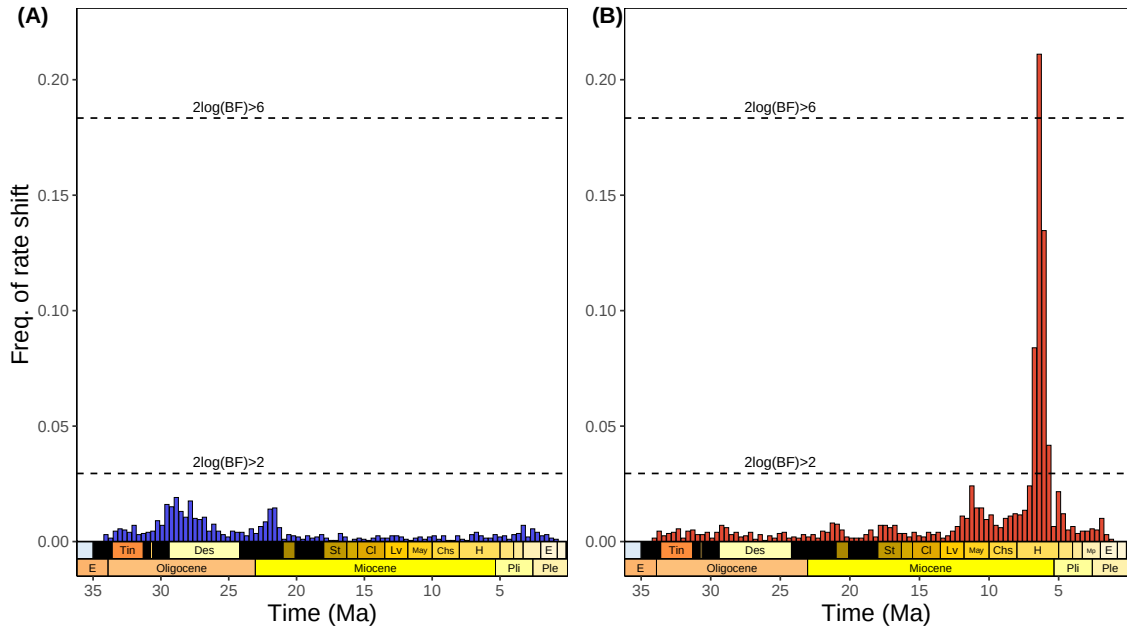


Figure S 6. Frequency of speciation (A) and extinction (B) rate shifts as estimated by the analysis of our species-level chinchilloid occurrence dataset with the rjMCMC model. Dashed horizontal lines indicate $2 * \log(\text{Bayes factor})$ of 2 and 6, respectively suggesting significant and strongly-supported rate shifts.

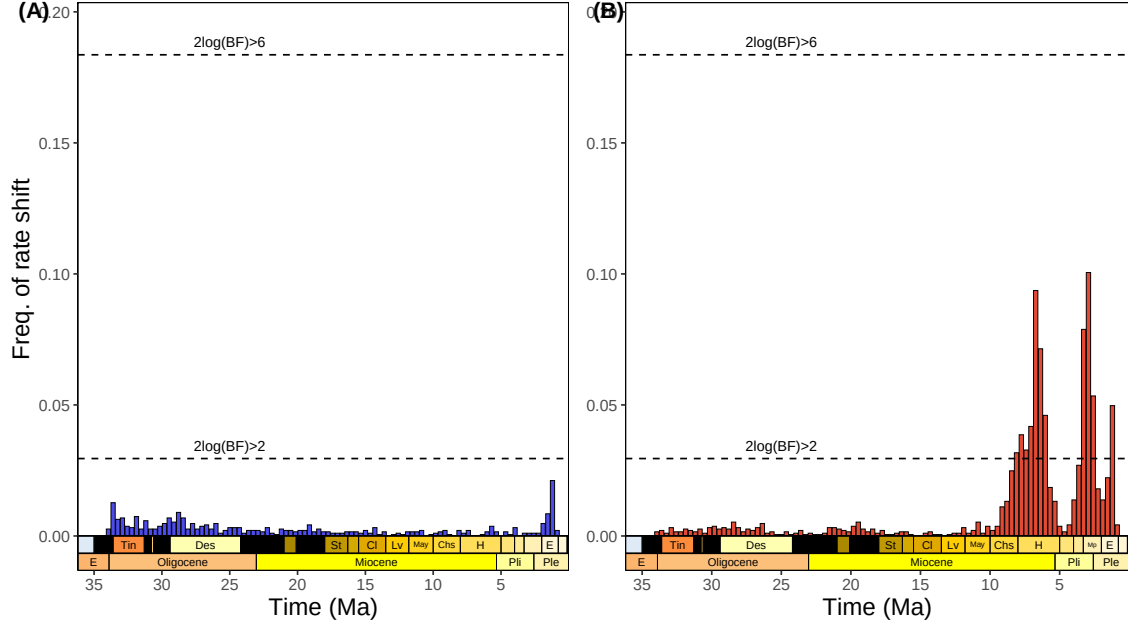


Figure S 7. Frequency of speciation (A) and extinction (B) rate shifts as estimated by the analysis of our genus-level chinchilloid occurrence dataset with the rjMCMC model. Dashed horizontal lines indicate $2 * \log(\text{Bayes factor})$ of 2 and 6, respectively suggesting significant and strongly-supported rate shifts.

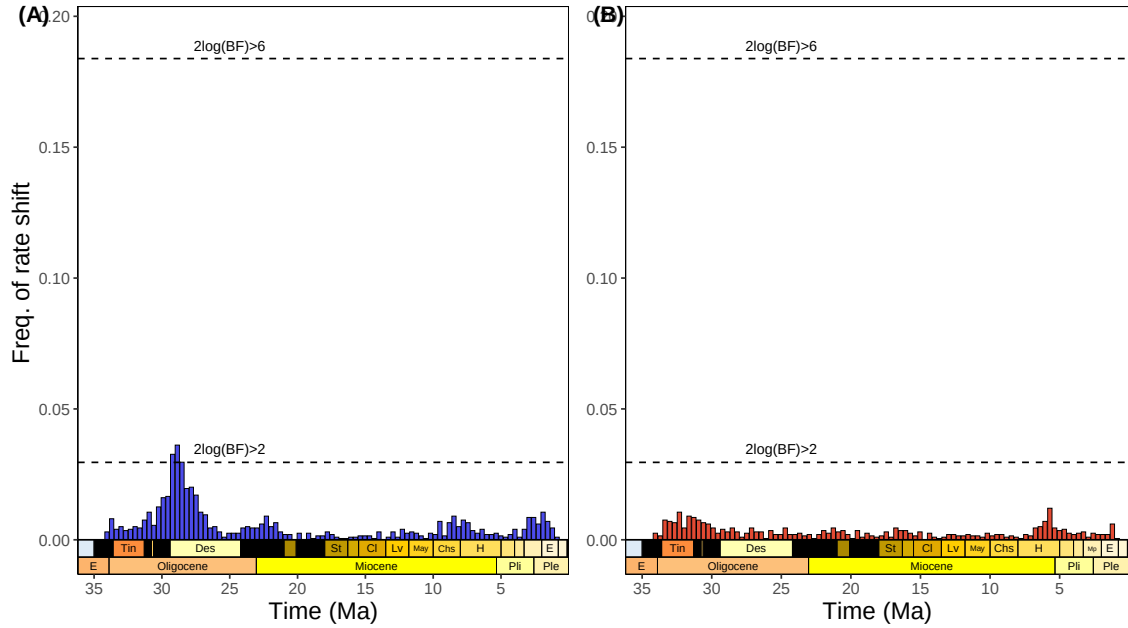


Figure S 8. Frequency of speciation (A) and extinction (B) rate shifts as estimated by the analysis of our species-level chinchilloid occurrence dataset with the rjMCMC model after removing singletons. Dashed horizontal lines indicate $2 * \log(\text{Bayes factor})$ of 2 and 6, respectively suggesting significant and strongly-supported rate shifts.

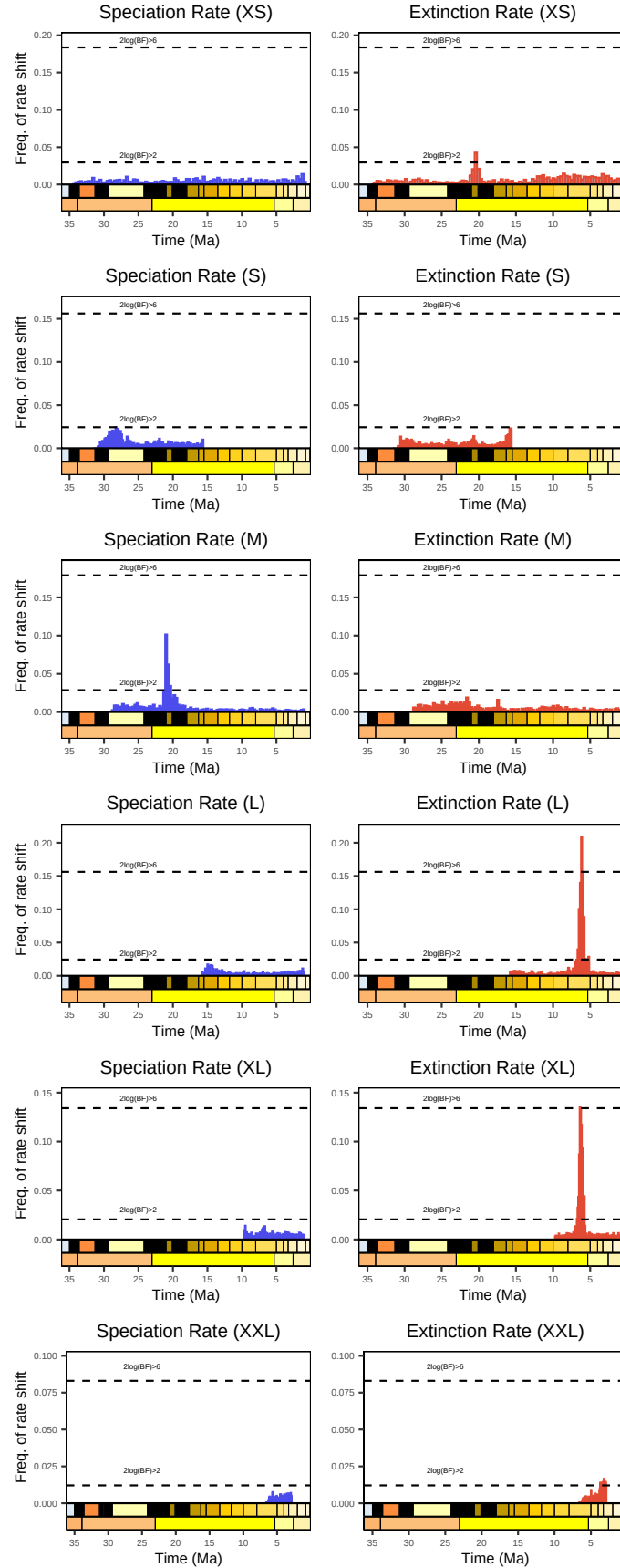


Figure S 9. Frequency of speciation and extinction rate shifts per body size class as estimated by the analysis of our species-level chinchilloid occurrence dataset with the rjMCMC model. Dashed horizontal lines indicate $2 \log(\text{Bayes factor})$ of 2 and 6, respectively suggesting significant and strongly-supported rate shifts.

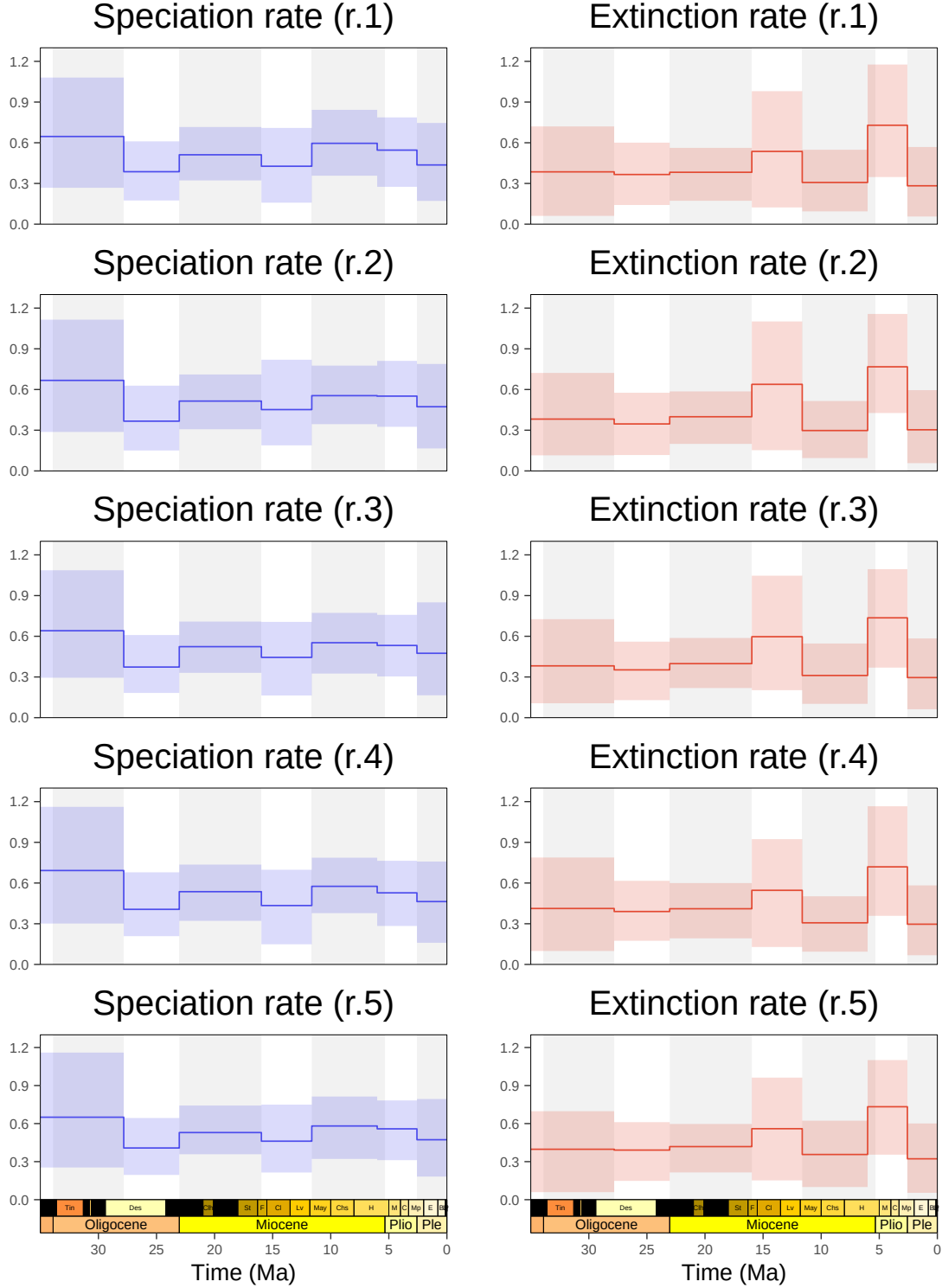


Figure S 10. Marginal chinchilloid speciation (blue) and extinction (red) rates through time as estimated by our BDNN analyses of the datasets with random trait (*i.e.*, body mass and/or hypsodonty index) imputation for taxa with missing trait reconstruction ($n = 11$). Caribbean taxa were excluded. Full lines show median rate estimates and ribbons their associated 95% Highest Posterior Density (HPD) interval, assessed across the 10 independent BDNN runs randomising fossil ages under a uniform law made for each trait imputation. This panel only depicts the rates through time from replicate 1 to 5.

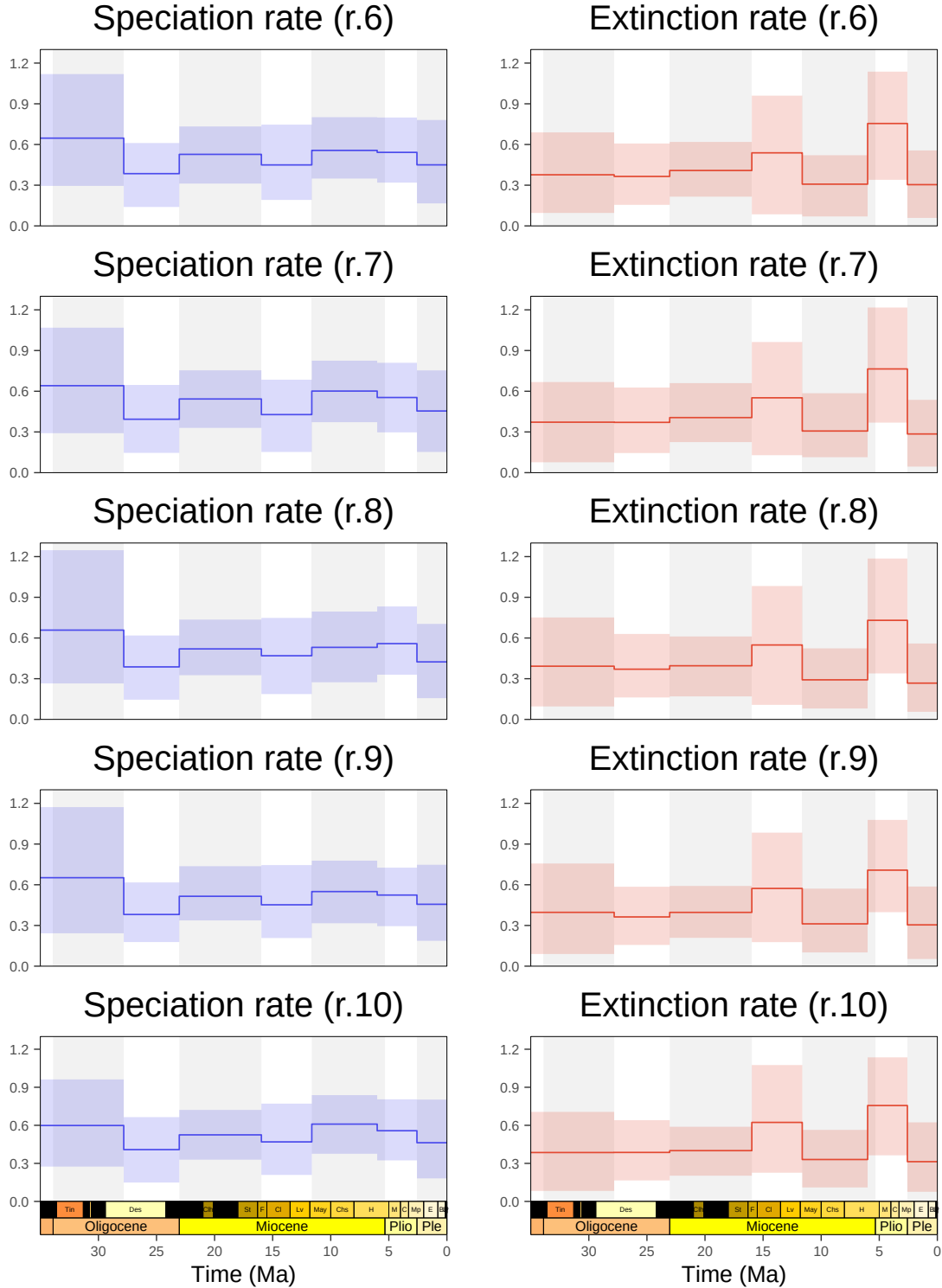


Figure S 11. Marginal chinchilloid speciation (blue) and extinction (red) rates through time as estimated by our BDNN analyses of the datasets with random trait (*i.e.*, body mass and/or hypsodonty index) imputation for taxa with missing trait reconstruction ($n = 11$). Caribbean taxa were excluded. Full lines show median rate estimates and ribbons their associated 95% Highest Posterior Density (HPD) interval, assessed across the 10 independent BDNN runs randomising fossil ages under a uniform law made for each trait imputation. This panel only depicts the rates through time from replicate 1 to 5.

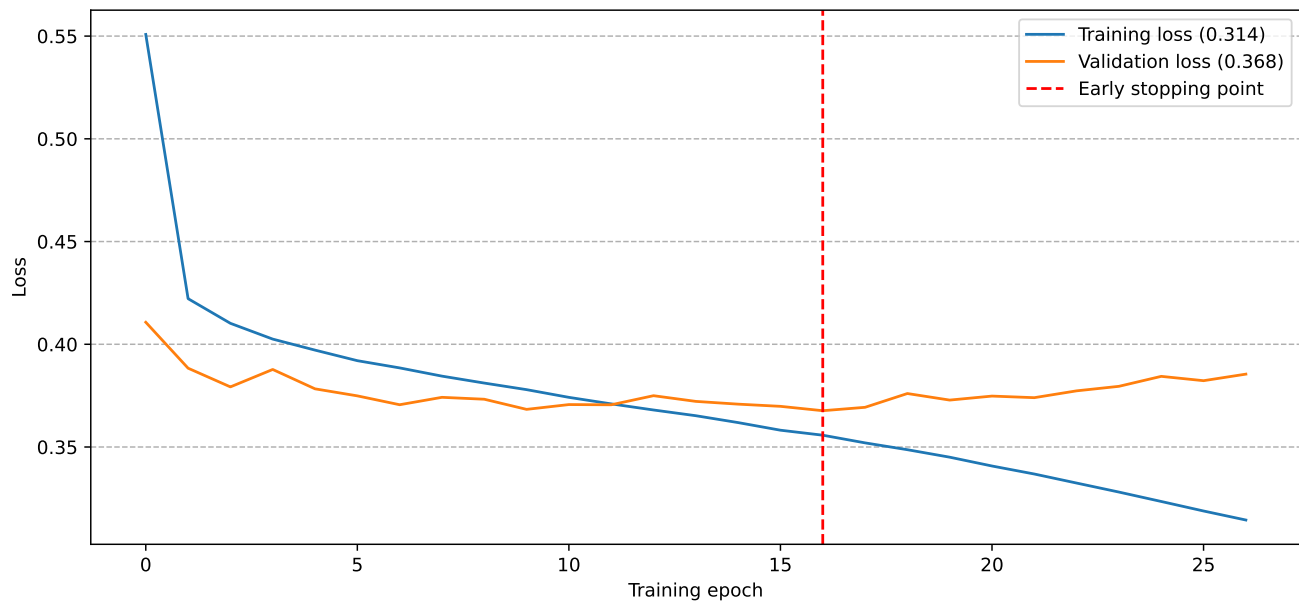


Figure S 12. Training and validation loss across training epochs for a DeepDive network comprising two LSTM units (64 and 32 nodes) connected to two dense layers (64 and 32 nodes) before the output layer. The red vertical line indicates the “early stopping point”, *i.e.*, when the validation loss starts increasing, indicative of overfitting. It determines the optimal value of number of iterations, at which we derive the training and validation loss values depicted between parenthesis.

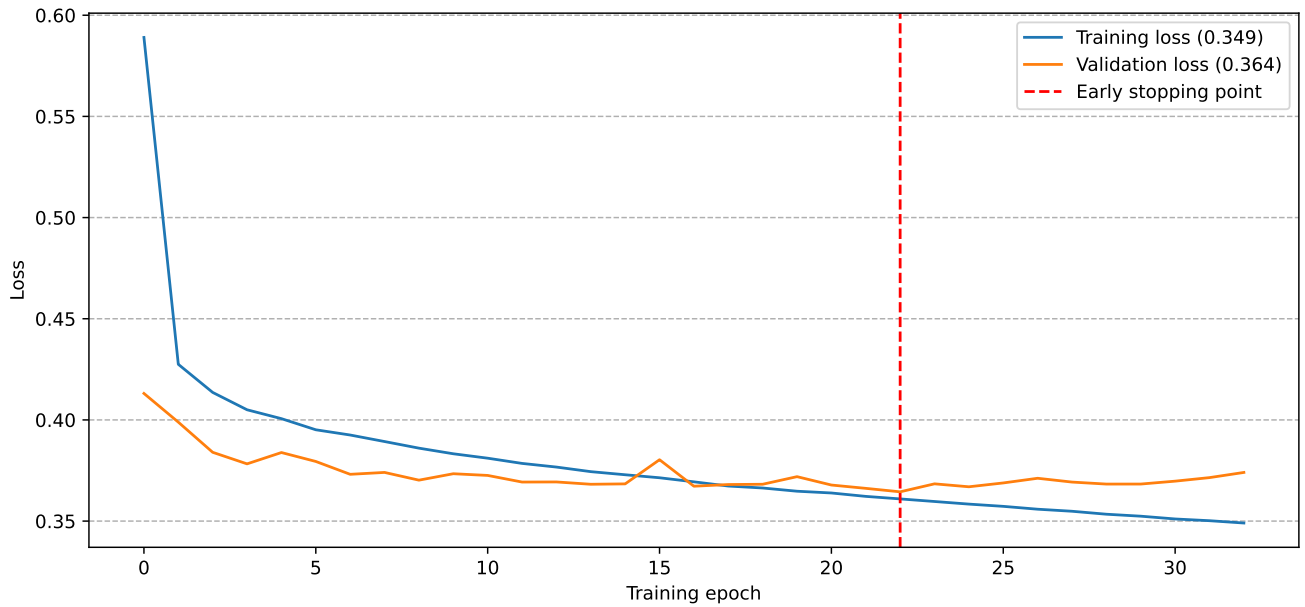


Figure S 13. Training and validation loss across training epochs for a DeepDive network comprising three LSTM units (128, 64 and 32 nodes) connected to two dense layers (64 and 32 nodes) before the output layer. The red vertical line indicates the “early stopping point”, *i.e.*, when the validation loss starts increasing, indicative of overfitting. It determines the optimal value of number of iterations, at which we derive the training and validation loss values depicted between parenthesis.

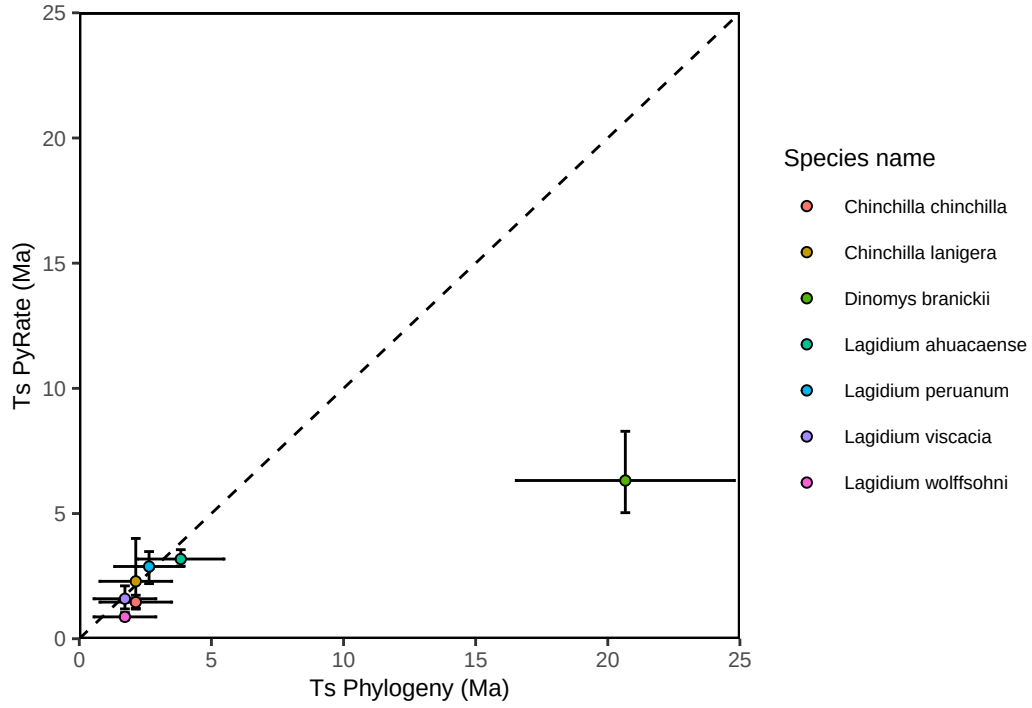


Figure S 14. Correspondance between divergence times leading to extant species estimated from our phylogeny ('Ts Phylo') and the times of species origination de novo estimated ('Ts PyRate') after implementing our sampling scheme for extant species absent from the fossil record (*Main text*, Fig. 1). Dots show mean estimates and error bars the associated 95% HPD. Dashed line shows the line with equation $y = x$.

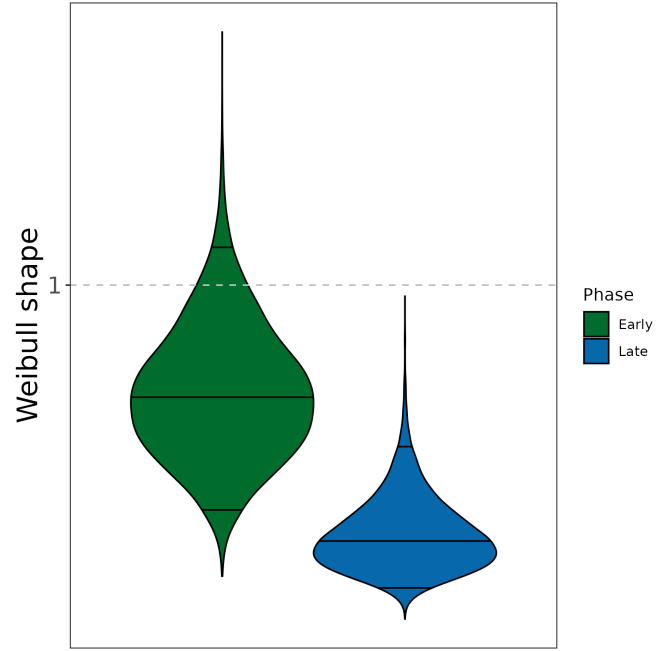


Figure S 15. Posterior estimate of the shape parameter of the Weibull distribution for the ADE analyses of the ‘Early’ (> 6 Ma) and ‘Late’ (< 6 Ma) occurrence datasets, each with constant extinction rate (Figure S6). 2.5%, 50% and 97.5% quantiles are indicated in the distribution of each parameter by black horizontal lines, respectively from bottom to top.

Supplementary Tables

Table S 1. Performance of the different network architectures tested in our DeepDive analyses. **LSTM Layers** and **Fully-connected** are respectively the number of Long Short Term Memory and Fully-connected units of the network (3: 128, 64 and 32 nodes; 2: 64 and 32 nodes; 1: 32 nodes). **Dropout** represents the Monte-Carlo dropout fraction, that is, the proportion of connections between neurons randomly toggled off between two learning epochs – one epoch being characterised by a complete pass through the entire training dataset –, preventing from overfitting. **Nb. training epochs** is the optimal number of training epochs. **Training loss** and **Validation loss** are the values of the Mean Squared Error (MSE) estimated between the ‘true’ diversity of each simulated dataset (resp. from the training or the validation set), and the predicted diversity made by the network after n rounds of optimisation, where n is the optimal number of training epochs. **Evaluation loss** represent the MSE between each true and its associated predicted diversity trajectory from the test dataset, once the is trained (*i.e.*, after n rounds of optimisation, where n is the optimal number of training epochs). The optimal number of epochs is determined when the **Validation loss** starts increasing (see Figure S12 and Figure S13). Architectures resulting in the smallest evaluation loss values, *i.e.*, generating the most accurate predictions, are depicted in bold.

LSTM layers	Fully-connected	Dropout	Nb. training epochs	Training loss	Validation loss	Evaluation loss
3	2	0.05	16	0.314	0.368	0.377
3	1	0.05	14	0.324	0.367	0.38
2	2	0.05	22	0.349	0.364	0.377
2	1	0.05	19	0.354	0.366	0.38
1	1	0.05	41	0.373	0.37	0.381

Table S 2. Table representing the lower and upper bounds of the 95% HPD (resp. ‘q_2.5%’ and ‘q_97.5%’) of the distribution of the divergence ages leading to all the extant chinchilloid species absent from the fossil record – *i.e.*, all species but *Lagostomus maximus*. Ages are expressed in Ma.

	<i>Chinchilla chinchilla</i>	<i>Chinchilla lanigera</i>	<i>Dinomys branickii</i>	<i>Lagidium ahuacaense</i>	<i>Lagidium peruanum</i>	<i>Lagidium viscacia</i>	<i>Lagidium wolffsohni</i>
q_2.5%	0.78	0.78	16.54	2.20	1.34	0.56	0.56
q_97.5%	3.50	3.50	24.79	5.48	3.94	2.90	2.90

Table S 3. References of the silhouettes used in Figure 1, from [Phylopic](#).

Silhouette	Author	License
<i>Chinchilla chinchilla</i>	Margot Michaud	CC0 1.0
<i>Lagidium peruanum</i>	Josue Hermosa	CC BY-SA 3.0
<i>Dinomys branickii</i>	Juan C. Cepeda Duque	CC BY 4.0
<i>Josephoartigasia monesi</i>	Nobu Tamura (body), Gustavo Lecuona (head)	CC BY-SA 3.0

Table S 4. Coefficient of speciation rate variation across the 10 BDNN replicates with random trait (*i.e.*, body mass and/or hypsodonty index) imputation for taxa with missing trait reconstruction ($n = 11$), excluding caribbean taxa. Simulated values (**cv_simulated**) were obtained from the analysis of 100 simulated datasets matching the empirical datasets in terms of number of traits, clade age and (approximately) number of extant species, generated under a constant-rate birth-death model. Empirical values (**cv_empirical**) were obtained from the analysis of the empirical dataset.

cv_empirical	cv_expected	Replicate
0.31	0.25	1
0.28	0.25	2
0.28	0.2	3
0.28	0.22	4
0.29	0.2	5
0.27	0.22	6
0.27	0.22	7
0.29	0.19	8
0.27	0.23	9
0.27	0.2	10
0.28	0.22	Total

Table S 5. Coefficient of extinction rate variation across the 10 BDNN replicates with random trait (*i.e.*, body mass and/or hypsodonty index) imputation for taxa with missing trait reconstruction ($n = 11$), excluding caribbean taxa.

cv_empirical	cv_expected	Replicate
0.55	0.44	1
0.51	0.44	2
0.5	0.37	3
0.49	0.32	4
0.48	0.27	5
0.5	0.33	6
0.54	0.46	7
0.54	0.29	8
0.48	0.38	9
0.48	0.41	10
0.51	0.37	Total

Table S 6. Predictor importance on speciation rate across the 10 BDNN replicates with random trait (*i.e.*, body mass and/or hypsodonty index) imputation for taxa with missing trait reconstruction ($n = 11$) after excluding caribbean taxa. **R. i** stands for replicate i, for i in [1,10]. The **SumRank** column indicates, for a given predictor, the sum of the ranks of this predictors across replicates. The **Rank** column represents the overall rank of the predictor (*i.e.*, sorting sum of the ranks by increasing order), with higher ranks (*i.e.*, closer to 1) indicating greater contribution to speciation rate variation across species. We highlighted in bold the three factors that contribute the most to variations in speciation rates across species.

Predictor	R. 1	R. 2	R. 3	R. 4	R. 5	R. 6	R. 7	R. 8	R. 9	R. 10	SumRank	Rank
Body_mass	1	1	1	1	1	1	1	1	1	1	10	1
Andean_uplift	2	3	2	2	2	2	2	3	2	2	22	2
Self_diversity	3	2	3	3	3	3	3	2	3	3	28	3
Tropicality	4	4	4	4	4	4	4	4	4	4	40	4
Sea_level	5	5	5	5	6	5	5	5	5	5	51	5
Hypsodonty	6	6	6	6	5	6	6	6	6	6	59	6
Temperature	7	7	7	7	7	7	7	7	7	7	70	7
time	8	8	8	8	8	8	8	8	8	8	80	8

Table S 7. Predictor importance on extinction rate across the 10 BDNN replicates with random trait (*i.e.*, body mass and/or hypsodonty index) imputation for taxa with missing trait reconstruction ($n = 11$) after excluding caribbean taxa. We highlighted in bold the three factors that contribute the most to variations in extinction rates across species.

Predictor	R. 1	R. 2	R. 3	R. 4	R. 5	R. 6	R. 7	R. 8	R. 9	R. 10	SumRank	Rank
Andean_uplift	2	1	1	2	2	1	1	1	1	1	13	1
Body_mass	1	2	2	1	1	2	2	2	2	2	17	2
Tropicality	3	3	4	4	4	3	3	3	4	3	34	3
Self_diversity	4	4	3	3	3	4	4	4	3	4	36	4
Hypsodonty	5	5	6	5	6	6	5	6	6	5	55	5.5
Temperature	6	6	5	6	5	5	6	5	5	6	55	5.5
Sea_level	7	7	7	7	7	7	7	7	7	7	70	7
time	8	8	8	8	8	8	8	8	8	8	80	8

Table S 8. Magnitude of the change in extinction rate (*i.e.*, **Fold Change**) between the two extreme body size classes (XS and XXL) across the 10 trait-imputed replicates without Caribbean taxa. **Upper CI** and **Lower CI** respectively show the upper and lower boundaries of the associated 95% HPD interval. The ‘Total’ row shows the averaged values across the 10 replicates.

Replicate	Fold Change	Upper CI	Lower CI
1	2.88	14.09	-3.23
2	2.38	10.75	-3.29
3	2.2	7.83	-3.33
4	2.62	11.09	-3.31
5	2.27	9.77	-2.68
6	2.06	8.11	-4.28
7	2.23	13.04	-3.97
8	2.29	13.17	-2.93
9	2.32	10.22	-2.53
10	2.43	9.87	-2.84
Total	2.37	10.79	-3.24

Table S 9. Magnitude of the change in extinction rate (*i.e.*, fold change) between tropical and extratropical taxa across the 10 trait-imputed replicates without Caribbean taxa. **Upper CI** and **Lower CI** respectively show the upper and lower boundaries of the associated 95% HPD interval. The ‘Total’ row shows the averaged values across the 10 replicates.

Replicate	Fold Change	Upper CI	Lower CI
1	1.62	4.11	-1.72
2	1.6	4.77	-1.5
3	1.56	3.98	-1.4
4	1.66	4.93	-1.51
5	1.42	3.36	-1.46
6	1.58	4.54	-1.47
7	1.64	5.44	-1.51
8	1.72	5.42	-1.6
9	1.55	4.27	-1.57
10	1.63	4.58	-1.48
Total	1.6	4.54	-1.52

Supplementary references

- Arnal, M., M. E. Pérez, and C. Deschamps. 2019. [Revision of the Miocene Caviomorph Rodents From the Río Santa Cruz \(Argentinean Patagonia\)](#). Publicación Electrónica de la Asociación Paleontológica Argentina 19.
- Bostelmann, J. E., J. P. L. Roux, A. Vásquez, N. M. Gutiérrez, J. L. Oyarzún, C. Carreño, T. Torres, R. Otero, A. Llanos, C. M. Fanning, and F. Hervé. 2013. [Burdigalian deposits of the Santa Cruz Formation in the Sierra Baguales, Austral \(Magallanes\) Basin: Age, depositional environment and vertebrate fossils](#). Andean Geology 40:458–489.
- Brandoni, D., L. G. Ruiz, M. F. Tejedor, G. Martin, and J. G. Fleagle. 2016. [Megatherioidea \(Mammalia, Xenarthra, Tardigrada\) from the Pinturas Formation \(Early Miocene\), Santa Cruz Province \(Argentina\) and their chronological implications](#). PalZ 90:619–628.
- Bucher, J., M. E. Pérez, L. R. González Ruiz, L. D'Elía, and A. Bilmes. 2021. [New middle miocene \(langhian - serravallian\) vertebrate localities in northwestern patagonia, argentina: A contribution to high latitude south american land mammal ages sequence](#). Journal of South American Earth Sciences 107:103024.
- Busker, F., M. E. Pérez, and M. T. Dozo. 2019. [A new chinchilloid \(rodentia, hystricognathi\) from the early miocene of the localities of bryn gwyn and gran barranca \(patagonia, argentina\)](#). Comptes Rendus Palevol 18:525–540.
- Esteban, G. I., S. M. Georgieff, N. L. Nasif, L. Ibáñez, and R. A. Bonini. 2019. Marco estratigráfico, paleontología de vertebrados y paleoambientes del plioceno de catamarca, tucumán y santiago del estero, argentina. Pp. 348–366 *in*.
- Esteban, G. I., N. L. Nasif, and S. M. Georgieff. 2014. Cronobioestratigrafía del mioceno tardio-plioceno temprano, puerta de corral quemado y villavil, provincia de catamarca, argentina. Acta geológica lilloan 26:165–192.
- Kramarz, A. 2002. [Roedores chinchilloideos \(Hystricognathi\) de la Formación Pinturas, Mioceno temprano-medio de la provincia de Santa Cruz, Argentina](#). Revista del Museo Argentino de Ciencias Naturales nueva serie 4:167–180.
- Kramarz, A. G. 2006. [Eocardiids \(rodentia, hystricognathi\) from the pinturas formation, late early miocene of patagonia, argentina](#). Journal of Vertebrate Paleontology 26:770–778.
- Kramarz, A. G. 2004. [Octodontoids and erethizontoids \(Rodentia, Hystricognathi\) from the Pinturas Formation, Early - Middle Miocene of Patagonia, Argentina](#). Ameghiniana 41:199–216.
- McGrath, A. J., J. Chick, D. A. Croft, H. E. Dodson, J. J. Flynn, and A. R. Wyss. 2022. [Cavioids, chinchilloids, and erethizontoids \(hystricognathi, rodentia, mammalia\) of the early miocene pampa castillo fauna, chile](#). American Museum Novitates 2022:1–46.
- Rasia, L. L. 2024. [Fossil record of lagostomus: A 10 million years story](#). Pp. 27–42 *in* L. L. Rasia, C. G. Barbeito, and F. Acuña, eds. Plains vizcachas: Biology and evolution of a peculiar neotropical caviomorph rodent. Springer Nature Switzerland, Cham.
- Rasia, L. L. 2016. [Los Chinchillidae \(Rodentia, Caviomorpha\) fósiles de la República Argentina: sistemática, historia evolutiva y biogeográfica, significado bioestratigráfico y paleoambiental](#).
- Rasia, L. L., and A. M. Candela. 2012. [Systematic and biostratigraphic significance of a chinchillid rodent from the pliocene of eastern argentina](#). Acta Palaeontologica Polonica 58:241–254.
- Rasia, L. L., M. de Los Reyes, S. Lorenzini, and A. M. Candela. 2025. [New insights on telicomys giganteus, the last giant dinomyid rodent from the pliocene of central argentina](#). Historical Biology 0:1–18.
- Ubilla, M., and A. Rinderknecht. 2016. [Lagostomus maximus \(desmarest\) \(rodentia, chinchillidae\), the extant plains vizcacha in the late pleistocene of uruguay](#). Alcheringa: An Australasian Journal of Palaeontology 40:354–365.
- Vizcaino, S. F., M. S. Bargo, M. E. Pérez, I. Aramendía, J. I. Cuitiño, E. S. Monsalvo, E. Vlachos, J. I. Noriega, and R. F. Kay. 2022. [Fossil vertebrates of the early-middle Miocene Cerro Boleadoras Formation, northwestern Santa Cruz Province, Patagonia, Argentina](#). Andean Geology 49:382–422.